

5 August 2002

TO: Pre-Installation Distribution Tab

SUBJECT: **DIRECTION 2142460, SIGNA LX UPGRADE PRE-INSTALLATION, REV 9**

Enclosed is *Direction 2142460, Signa LX Upgrade Pre-Installation*, Rev 9 update. Table 1 below summarizes the major information incorporated in this revision. Changed content on individual pages is indicated by revision bar in the left hand margin.

TABLE 1
DIRECTION 2142460 REVISION 9 MAJOR CHANGES INCORPORATED

TAB	MAJOR INFORMATION CHANGES
1 SYSTEM CONFIGURATION	●Sec 1-3 flowcharts updated for M3033KH, M3033KJ, & M3033KK ●Corrected 1.0T CX Magnet listing in flowchart Illustration 1-20
4 SITE ENVIRONMENT	●Revised system acoustic noise levels
10 TOOLS & TEST EQUIPMENT	●Added Resistance Meter to table of Rigger/Customer Supplied Equipment

UPDATE INSTRUCTIONS FOR YOUR DIRECTION 2142460 REV 8

Refer to Table 2 for instructions to properly update your Direction 2142460 Rev 8 with the enclosed Revision 9 pages.

TABLE 2
DIRECTION 2142460 REVISION 8 UPDATE INSTRUCTIONS

TAB	REMOVE PAGE(S)	INSERT PAGE(S)
Front Matter	Title Page	Title Page
	A/B	A/B
	i to iv	i to iv
1 SYSTEM CONFIGURATION	1-1/2 1-27 to 1-50	1-1/2 1-27 to 1-51
4 SITE ENVIRONMENT	4-7/8	4-7/8
10 TOOLS & TEST EQUIPMENT	10-1 to 10-10 (entire section)	10-1 to 10-11 (entire section)

Debra C. Balis
MR SERVICE ENGINEERING



GE Medical Systems

Technical Publications

Direction 2142460

Revision 9

Signa® LX Upgrade Pre-Installation

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Operating Documentation



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WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTE AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

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REVISION HISTORY

REV	DATE	PRIMARY REASON FOR CHANGE
Rev 0	... Oct. 31, 95	... Initial release of document.
Rev 1	 Dec. 6, 96	... Added Signa Horizon LX information throughout manual. Incorporated miscellaneous corrections.
Rev 2	 Jan. 9, 98	... Added Signa Horizon LX SGD information throughout manual. Incorporated miscellaneous corrections. All pages revised due to text font change required for CD-ROM and web page compatibility of this document.
Rev 3	... Sep. 18, 98	... Added Signa CVMR upgrades information throughout manual. Revised Cx magnet ground wires connect to single ground stud. Vibration specification revised for clarity and Appendix A revised to enhance vibration guidelines information. Incorporated miscellaneous corrections.
Rev 4	... Apr. 30, 99	... Revised Recon Processor catalogs to new Reconstruction Module catalogs. Updated Signa CV/i upgrade catalogs. RF/Pen Cabinet front mounting hole height dimension changed. Operator Workspace Table illustration added note for integral part of system. Added dimension illustrations for equipment position on top of Operator Workspace Table. Revised Heat Output values. Revised Facility Input Power Requirements. Updated tools lists, added new items and deleted obsolete items.
Rev 5	 Aug. 8, 00	... Incorporated Release 8.3 software/hardware catalogs (i.e. UCERD). Revised Signa CVMR information for Release 8.4 (ADW to AW). Deleted discontinued Horizon Upgrade catalogs information. Revised Spectroscopy Upgrade/Options information & added MNS Cabinet throughout the manual. Revised Heat Output values. Revised Power Distribution System information. Updated Interconnect Data new catalogs & discontinued catalogs. Clarified RF Penetration Panel upgrade information for RF Shielded Room. Updated references to Upgrade Ordering Checklist documents. Section 10 Table Of Contents corrections.
Rev 6	 Feb. 6, 01	... Changed document title to Signa LX Upgrade Pre-Installation. Incorporated Release 8.X ASP1 Release hardware (i.e. ACGD/PDU Cabinet & SRF Cabinet) information throughout the manual. Revised Signa CVMR information for Release ASP1. Deleted discontinued Horizon Upgrade catalogs information. Revised Spectroscopy Upgrade/Options information. Changed vibration values to G units of measure at Field's request. Updated Interconnect Data new catalogs & discontinued catalogs.
Rev 7	 Mar. 7, 01	... Sec 1 added WARNING for system upgrade to ACGD/PDU to check facility power, MDC, and full system UPS (if present) compatibility with ACGD system config power requirements. Sec 5 updated Critical Power Requirements max. momentary demand for revised design specifications; expanded WARNING to check full system UPS (if present); added 5-2-2 Power Demands Duration & Regulation to clarify power requirements for Signa system with ACGD.
Rev 8	 June 3, 02	... Replaced Direction 2128126 Rev 0 with Rev 1. Added 1.5T Signa MR/i with EXCITE Technology system upgrade information, Added Upgrade to CXK4 Magnet information/references, and deleted Signa CVMR Upgrade information throughout the document. Other major changes listed by section include the following Sec 1: Throughout section added note for systems with Oxford & S-I magnet must upgrade to LCC Magnet; Added 1.5T Signa MR/i with EXCITE Technology catalogs; Deleted OW Color Monitor information; Updated Signa LX Upgrade Reconstruction Module & BAM Memory catalogs; Deleted Oxford & S-1 Magnet related upgrade catalogs which are no longer PriceBook listed; Deleted Signa CVMR related upgrade catalogs which are no longer PriceBook listed. Sec 2: Updated Gradient Coil Assembly, cradle & cart dimensions; Added Network connection and telephone lines option to system connectivity requirements. Sec 3: Changed Signa Horizon to Signa LX. Sec 4: Revised Magnet Room temperature range per System Specifications; Added upgrade to LCC Magnet information references; Deleted OW Color Monitor heat output; Updated Magnet Room Temperature & Humidity Specifications to match system specifications; Updated System Acoustic Noise Level Operating Conditions & wall construction examples. Sec 5: Revised Voltage Transient specification; Added NEC 2002 Article references; Update Critical Power Requirements; Added Ground Fault Protection information. Sec 6: Added 1.5T Signa MR/i with EXCITE Technology information; Updated miscellaneous system interconnects. Sec 7: Added Upgrade to CXK4 Magnet references. Sec 9: Updated SOI Request Form web address; Added SAFETY subsection; Added Upgrade to CXK4 Magnet subsections. Sec 10: Added new & deleted obsolete test equipment & part numbers.
Rev 9	.. Aug. 5, 2002	... Sec 1: Sec 1-3 flowcharts updated for M3033KH, M3033KJ, & M3033KK; Corrected 1.0T CX Magnet listing in flowchart Illustration 1-20. Sec 4: Revised system acoustic noise levels. Sec 10: Added Resistance Meter to table of Rigger/Customer Supplied Equipment.

REVISION HISTORY

<u>PAGE</u>	<u>REV</u>	<u>PAGE</u>	<u>REV</u>	<u>PAGE</u>	<u>REV</u>	<u>PAGE</u>	<u>REV</u>	<u>PAGE</u>	<u>REV</u>
Title Page	9	2-1	6	4-3	8	6-1 to 6-5	8	PREINSTALL.	
GE Logo	—	2-2	blank	4-4	6	6-6	6	CHECKLIST	
Direction 2128126 ...	1*	2-3 to 2-7	8	4-5	8	6-7 to 6-58	8	Tab MR259	—
A to B	8	2-8 to 2-9	6	4-6 to 4-7	6			9-1	8
i	9	2-10	8	4-8	9	RF SHIELDED ROOM		9-2	blank
ii to iii	8	2-11 to 2-12	6	4-9 to 4-15	8	Tab MR257	—	9-3 to 9-7	8
iv	9	2-13 to 2-26	8			7-1	8		
v	7	2-27 to 2-51	9	POWER		7-2	blank	TOOLS & TEST	
SYSTEM CONFIG.		MAGNETIC FIELD		REQUIREMENTS		7-3 to 7-5	8	EQUIPMENT	
Tab MR251	—	CONSID.		Tab MR255	—			Tab MR260	—
1-1	9	Tab MR253	—	5-1	8	SHIPPING &		10-1	9
1-2	blank	3-1	8	5-2	blank	DELIVERY		10-2	blank
1-3 to 1-26	8			5-3 to 5-9	8	Tab MR258	—	10-3 to 10-11	9
1-27 to 1-51	9	SITE ENVIRONMENT		5-10 to 5-13	7	8-1	6		
		Tab MR254	—	5-14	8	8-2	blank	A	
ROOM LAYOUTS		4-1	8	INTERCONNECT DATA		8-3	8	Tab MR301	—
Tab MR252	—	4-2	blank	Tab MR256	—	8-4 to 8-26	6	A-1 to A-7	6

* This revision number/letter corresponds to the indicated document's revision control system.

TABLE OF CONTENTS

<u>SECTION</u>	<u>PAGE</u>
TABLE OF CONTENTS	
INTRODUCTION	iv
SECTION 1 – SYSTEM CONFIGURATION	
1-1 INTRODUCTION	1-3
1-2 UPGRADES TO 1.5T SIGNA LX WITH EXCITE TECHNOLOGY	1-4
1-2-1 Signa Horizon LX	1-6
1-2-2 Signa Horizon 5.X System	1-9
1-2-3 Signa Advantage 1.5T 5.X & 4.X	1-14
1-3 UPGRADES TO SIGNA HORIZON LX WITH ACGD	1-22
1-3-1 Signa Horizon LX with SGD & Signa MR/i System	1-24
1-3-2 Signa Horizon LX With 8645 Gradients System	1-27
1-3-3 Signa Horizon 5.X System	1-29
1-3-4 Signa Horizon 5.X Software Only & Signa Advantage 5.X System	1-35
1-3-5 Signa Advantage 4.X System	1-41
1-4 SIGNA HORIZON LX SYSTEM OPTIONS	1-46
1-5 UPGRADES TO SIGNA HORIZON LX WITH SGD SYSTEM	1-49
1-6 SIGNA HORIZON 5.5 SYSTEM OPTIONS	1-49
1-7 TRANSPORTABLE/RELOCATABLE UPGRADE	1-50
1-8 TRUE MOBILE UPGRADE	1-50
1-9 VAN MANUFACTURERS CONTACT INFORMATION	1-51
SECTION 2 – ROOM LAYOUTS	
2-1 INTRODUCTION	2-3
2-2 TWO MAGNET SITE LAYOUT	2-4
2-3 MINIMUM DOOR/HALLWAY SIZES	2-4
2-3-1 Gradient Coil Assembly, Cradle, & Cart Requirements	2-6
2-4 CABLING CONSIDERATIONS	2-8
2-5 SPECIAL SITING CONSIDERATIONS	2-12
2-5-1 Power Distribution Unit (PDU)	2-12
2-5-2 Blower Box (MG6)	2-13
2-5-3 Water Chiller (WC1) For Gradient Coil Cooling	2-13
2-5-4 SPT Phantom Set Shipping/Storage Cabinet	2-13
2-5-5 Spectroscopy for 1.5T Magnet Only (Optional)	2-14
2-5-6 Magnet Monitor	2-14
2-5-7 System Monitoring & Support Connectivity Requirements	2-15
2-6 ARCHITECTURAL REMINDERS	2-15
2-7 FLOOR LOADING AND WEIGHTS	2-16
2-7-1 Operator Workspace Mounting Requirements For Signa LX	2-17
2-8 COMPONENT DIMENSIONS	2-18
SECTION 3 – MAGNETIC FIELD CONSIDERATIONS	3-1

<u>SECTION</u>	<u>PAGE</u>
SECTION 4 – SITE ENVIRONMENT	
4-1 INTRODUCTION	4-3
4-2 TEMPERATURE AND HUMIDITY SPECIFICATIONS	4-3
4-3 COOLING REQUIREMENTS	4-4
4-3-1 Air Cooling	4-4
4-3-2 Water Cooling	4-6
4-4 ALTITUDE	4-6
4-5 LIGHTING	4-6
4-6 ACOUSTICS	4-7
4-7 ROOM VENTILATION	4-11
4-8 CRYOGENIC VENTING	4-11
4-9 LCC (ACTIVE SHIELD) MAGNET CHANGING MAGNETIC ENVIRONMENT SPECIFICATIONS	4-12
4-10 CONSTRUCTION MATERIALS	4-12
4-11 VIBRATION	4-12
4-11-1 Definition	4-12
4-11-2 Types Of Vibration Image Quality Issues	4-12
4-11-3 Specifications For 1.5T S-II, & S-III (Unshielded) Magnets and 1.5T S-III Magnet With Magnishield	4-13
4-11-4 Specifications For 1.5T S-IV, S-V, & CX And 1.0T S-X & CX Magnets ..	4-14
4-11-5 Specifications For 1.5T And 1.0T LCC Magnets Without Gradient Isolation	4-14
4-11-6 Specifications For 1.5T And 1.0T LCC Magnets With Gradient Isolation ..	4-15
SECTION 5 – POWER REQUIREMENTS	
5-1 INTRODUCTION	5-3
5-2 CRITICAL POWER REQUIREMENTS	5-4
5-3 POWER DISTRIBUTION SYSTEM	5-6
5-3-1 Main Disconnect Control (MDC)	5-6
5-3-2 System Power Distribution Unit	5-9
5-4 GROUNDING	5-10
5-4-1 Facility Ground	5-10
5-4-2 System Ground	5-12
5-5 GROUND FAULT PROTECTION	5-14
SECTION 6 – INTERCONNECT DATA	
6-1 INTRODUCTION	6-3
6-1-1 Component Designators	6-4
6-1-2 Definition of Terms	6-5
6-2 SYSTEM INTERCONNECTS ADDED BY UPGRADES	6-6
6-2-1 Introduction	6-6
6-2-2 M1033L – Indigo to Octane II with EXCITE Upgrade	6-7
6-2-3 M1060JW – Shield/Cryo Cooler Compressor	6-7
6-2-4 M1060LC – 1.5T CXK4 Magnet Upgrade with Wide Open Enclosure	6-8
6-2-5 M1090JP & M1090JR – Enclosure Upgrade For Fixed Site	6-8
6-2-6 M1090JW – Laser to Halogen Alignment Lights Upgrade	6-9
6-2-7 M1090PC – 5.5 Phased Array Compatibility Kits for 1.5T System	6-9
6-2-8 M1090PH – Pre LX to LX Without Phase Array To 4 channel Phase Array Adder Kit For 1.5T System	6-9
6-2-9 M1090PW – 1.5T Phased Array Upgrade Kit (UCERD) For LX Systems Without Phase Array	6-10
6-2-10 M1090SJ – 5.5 To LX Upgrade For Fixed Site	6-10
6-2-11 M1191JF – Enclosure Upgrade For Transportable/Relocatable	6-10

SECTION**PAGE****SECTION 6 – INTERCONNECT DATA (Continued)**

6-2-12	M1191JM – 5.5 To LX Cable Upgrade for True Mobile with Rear Mounted Penetration Panel	6-10
6-2-13	M1191JN – 5.5 To LX Cable Upgrade for Transportable/Relocatable (T/R) with Rear Mounted Penetration Panel	6-10
6-2-14	M1191SN – Collector Kit for Upgrade to 1.5T LCC Magnet	6-11
6-2-15	M1191SP – Collector Kit for Upgrade to 1.5T LCC Magnet	6-13
6-2-16	M1860LC – 1.0T CXK4 Magnet Upgrade with Wide Open Enclosure	6-13
6-2-17	M1890PA – 5.5 Phased Array Compatibility Kits For 1.0T System	6-13
6-2-18	M1890PH – Pre LX to LX Without Phase Array to 4 channel Phase Array Adder Kit For 1.0T System	6-13
6-2-19	M1890PW – 1.0T Phased Array Upgrade Kit (UCERD) for LX Systems without Phased Array	6-13
6-2-20	M1891SN – Collector Kit for Upgrade to 1.0T LCC Magnet	6-14
6-2-21	M1891SP – Collector Kit for Upgrade to 1.0T LCC Magnet	6-15
6-2-22	M3090BA – 1.0T ACGD Upgrade for system with SGD Gradient Cabinet & RF/Penetration Cabinet	6-16
6-2-23	M3090BB – ACGD Upgrade for system with SGD Gradient Cabinet & RF/PDU Cabinet	6-16
6-2-24	M3090BC – 1.5T ACGD Upgrade for system with SGD Gradient Cabinet & RF/Penetration Cabinet	6-16
6-2-25	M3090CA – 4.X To LX With ACGD Cable Upgrade For Fixed Site	6-17
6-2-26	M3090CB – 5.1–5.4 To LX With ACGD Cable Upgrade For Fixed Site ...	6-21
6-2-27	M3090CC – 5.5 Software Only Site To LX With ACGD Cable Upgrade For Fixed Site	6-23
6-2-28	M3090CD – 5.5 To LX With ACGD Cable Upgrade For Fixed Site	6-24
6-2-29	M3090CH – 3.X–5.4 to LX with ACGD Cable Upgrade for Transportable/Relocatable (T/R) with Rear Mounted Penetration Panel	6-25
6-2-30	M3090CJ – 5.5 Software Site to LX with ACGD Cable Upgrade for Transportable/Relocatable (T/R) with Rear Mounted Penetration Panel ..	6-25
6-2-32	M3090CM – 5.X To LX with ACGD Cable Upgrade for Fixed Site	6-28
6-2-33	M3090CN – 5.5 To LX Cable Upgrade for Transportable/Relocatable (T/R) with Rear Mounted Penetration Panel	6-30
6-2-34	M3090EB – 1.5T EXCITE Hardware Kit	6-30
6-3	ADVANTAGE WORKSTATION INTERCONNECTS	6-31
6-4	INTERCONNECTS RE-ROUTED BY HORIZON LX WITH ACGD UPGRADES	6-31
6-5	INTERCONNECTS REMOVED BY LX WITH EXCITE TECHNOLOGY UPGRADES	6-32
6-6	INTERCONNECTS REMOVED BY HORIZON LX WITH ACGD UPGRADES	6-34
6-6-1	Signa Horizon LX System With SGD & RF/PDU Cabinet	6-34
6-6-2	Signa Horizon LX System With SGD & RF/Penetration Cabinet	6-35
6-6-3	Signa Horizon LX System With 8645 Gradients & RF/Penetration Cabinet	6-37
6-6-4	Signa Horizon 5.X System	6-39
6-6-5	Signa Horizon 5.X Software & Signa Advantage 5.X System	6-42
6-7	INTERCONNECTS REMOVED BY SIGNA HORIZON 5.X TO SIGNA LX COMPUTER ONLY UPGRADE	6-52
6-8	REAR PEDESTAL AND ENCLOSURE CABLE ACCESS	6-54
6-9	SPECTROSCOPY OPTION INTERCONNECTS FOR 1.5T SYSTEM WITH ACGD ..	6-55
6-9-1	M3090DA – 4.X–5.4 MNS Upgrade Kit For ACGD Site	6-55
6-9-2	M3090DB – 5.5 MNS Upgrade Kit For ACGD Site	6-56
6-9-3	M3090DB – LX MNS Upgrade Kit For ACGD Site	6-56
6-10	SPECTROSCOPY OPTION INTERCONNECTS FOR 1.5T SYSTEM WITH SGD ..	6-57

<u>SECTION</u>	<u>PAGE</u>
SECTION 7 – RF SHIELDED ROOM	
7-1 RF SHIELDED ROOM SPECIFICATION	7-3
7-2 VENTS	7-3
7-3 PLUMBING	7-3
7-4 ELECTRICAL	7-3
7-5 RF PENETRATION PANEL	7-3
7-6 ANCHOR HARDWARE REQUIREMENTS FOR MR EQUIPMENT INSIDE RF SHIELDED ROOM	7-5
7-6-1 Magnet Mounting	7-5
7-6-2 Blower Box	7-5
7-7 MAGNET ROOM EQUIPMENT MOUNTING	7-5
7-7-1 Magnet Rundown Unit (MS4)	7-5
7-7-2 Emergency Off Buttons	7-5
7-8 ROOM VENTILATION SWITCH	7-5
SECTION 8 – SHIPPING AND DELIVERY DATA	
8-1 INTRODUCTION	8-3
8-2 UPGRADE CATALOGS SHIPPING INFORMATION	8-6
SECTION 9 – PREINSTALLATION CHECKLIST	
9-1 INTRODUCTION	9-3
9-2 PREPARATIONS REQUIRED IN ADVANCE OF UPGRADE DELIVERY	9-3
9-3 GENERAL PREINSTALLATION REMINDERS	9-5
9-4 SAFETY	9-5
9-5 PREPARATIONS REQUIRED IN ADVANCE OF UPGRADE LCC MAGNET DELIVERY AND MOVING INTO MAGNET ROOM	9-6
9-6 PREPARATIONS REQUIRED IN ADVANCE OF UPGRADE LCC MAGNET CRYOGEN TRANSFER/MAGNET FILL	9-7
9-7 PREPARATIONS REQUIRED IN ADVANCE OF UPGRADE LCC MAGNET RAMP-UP	9-7
SECTION 10 – TOOLS AND TEST EQUIPMENT	
10-1 INTRODUCTION	10-3
10-5 INSTALLATION EQUIPMENT	10-6
10-7 TEST EQUIPMENT	10-8
10-8 CALIBRATION TOOLS/FIXTURES	10-9
10-9 TOOL KIT	10-10
APPENDIX A – MR SITE VIBRATION TEST GUIDELINES	
A-1 TEST MEASUREMENTS	A-1
A-2 EQUIPMENT (SPECTRAL ANALYZER) SET-UP	A-1
A-3 TEST MEASUREMENTS	A-2
A-3-1 Ambient Baseline Condition	A-2
A-3-2 Normal (Operating) Condition	A-2
A-4 PRESENTATION/INTERPRETATION OF RESULTS	A-3

INTRODUCTION

This document contains the physical, magnetic, cryogenic, plumbing and electrical data necessary for planning and preparing a site for a Signa Horizon system upgrade. “Preinstallation work” is done to prepare the customer’s premises for the installation of the upgrade sold. It is the responsibility of the purchaser to arrange for performance of this work. Such work includes:

- Installation of the electrical conduit, junction boxes, ducts, surface raceways, outlets and line safety switches.
- Installation of wires not supplied by GE Medical System such as facility input power line to the Main Disconnect Control and Power Distribution Unit. If new wires are installed then the electrical contractor shall ring identify and tag all wires at both ends. Color-coded wires are recommended for easier identification. Wires shall be continuous without splices. Insulation on all equipment ground wires must be green with a yellow stripe.
- Installation of non-electrical lines such as: water plumbing, helium venting systems and air conditioning equipment. Also, installation of recommended air, vacuum and oxygen lines into the Magnet Room. All lines must be clearly labeled.
- Site renovation.
- Installation of structural reinforcements as required.

All work **MUST** be in compliance with national and local building and safety codes.

All site plans and preliminary concepts should be reviewed by GE Medical Systems MR Siting group prior to construction.

Unless specifically mentioned, GE Medical Systems does not provide or install: the facility input power lines to the Main Disconnect Control and Power Distribution Unit or the power lines required in the Magnet Room, the Main Disconnect Control option, nor raised flooring, conduit, junction boxes, ducts, plumbing, water treatment equipment, cryogenic venting outside the Magnet Room, non standard cryogenic venting inside the Magnet Room, or the RF shielded room specified in this document.

All electrical installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations, and testing shall be performed by qualified GE Medical Systems personnel or third-party service companies with equivalent training. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE’s electrical work on these products will comply with the requirements of the applicable electrical codes. The purchaser of GE equipment shall only utilize qualified personnel (i.e. GE’s field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

Pre-installation information is continually changing due to the evolution of the product. GE will make every reasonable effort to maintain accuracy of the pre-installation information.